

FACTSHEET FS04 · Low volatility

What changed on 2 October 2026. Two corrections, both to the universe of stocks, and every number below is re-run on them. (1) **US:** the universe is now the actual S&P 500 members at each month-end. Until now it was the 500 largest of every company that was *ever* in the index 2012–2026, which includes later winners before they joined. (2) **UK and EU:** the price panels now include the index members that stopped trading (UK coverage of members 71% → 92%, EU 81% → 89%); before, most of them had no prices. SCANDI stays as registered (OMXC25 + OMXS30 + OMXH25). Registered verdicts are unchanged; where a corrected number crosses a gate, the text says so. Comparison and method: FACTSHEETS_RERUN_2026-10-02.md.

Do calm stocks beat exciting ones?

Henrik Gjerning · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY	ORIGIN	FAMILY
Defensive / low-risk anomaly	Haugen & Heins (1975); Blitz & van Vliet (2007)	Low risk
UNIVERSES	BACKTEST	BOOKS
US · EU · UK · DK · SCANDI · World (point-in-time)	Feb 2013 – Aug 2026, monthly, net of costs	Long/short and long-only (calmest stocks)
REBALANCE	STATUS	
Monthly	Descriptive factsheet, not a registered trial	

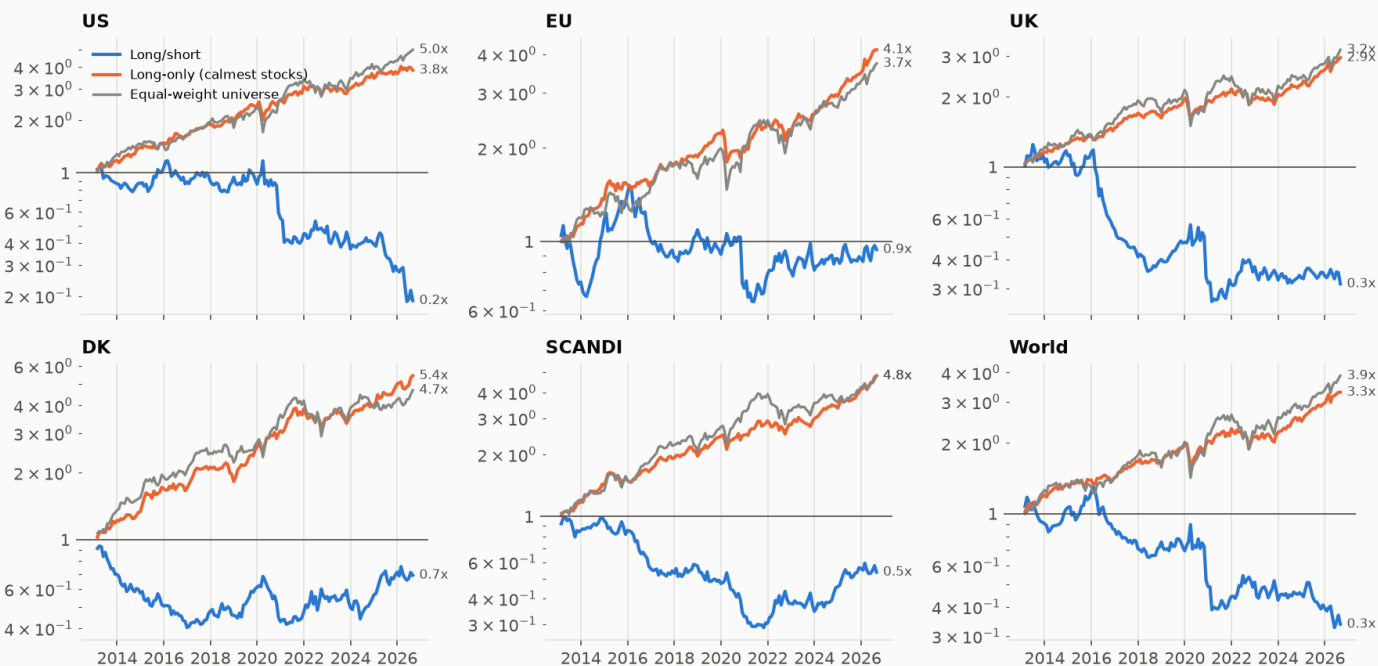
The claim. Low-volatility stocks earn about the same or more than high-volatility stocks, so their risk-adjusted return is far higher, contradicting the CAPM. Blitz & van Vliet (2007) documented it globally for 1986–2006; Ang, Hodrick, Xing & Zhang (2006) for idiosyncratic volatility. **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned -8.7% to +1.6% a year and was positive in 1 of the six; no universe passes the 2.87 gate. The long-only book (calmest stocks) had a higher Sharpe than the equal-weight universe in all six, with alphas of +2.7% to +5.0%, none past the gate. **What goes wrong (§10):** the main drags are a market bet (average beta -0.79, worth -10.0% a year in 2013–19). None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Low risk; its returns sit closest to the low risk cluster of the signal library. **360° view (§12):** nearest neighbours Volatility, Range (MAX–MIN); after its closest neighbours and the market it keeps an alpha of -3.0% to +5.4% (largest |t| 2.5), so it carries some information of its own.

1. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
US	502 / 10	-8.7%	-1.36	+4.6% (1.1)	-1.01	+10.4%	0.89	-21%	+12.6%	0.85	+14.9%
EU	264 / 10	+1.6%	0.28	+10.9% (2.4)	-0.87	+11.0%	1.07	-21%	+10.2%	0.75	+10.2%
UK	326 / 10	-6.0%	-0.96	+2.8% (0.6)	-0.92	+8.3%	0.93	-20%	+8.9%	0.68	+7.8%
DK	20 / 3	-1.1%	-0.21	+7.2% (1.8)	-0.66	+13.3%	1.07	-19%	+12.0%	0.83	+10.2%
SCANDI	70 / 5	-3.1%	-0.63	+6.1% (1.7)	-0.74	+12.3%	1.13	-15%	+12.3%	0.89	+11.0%
World	1088 / 10	-5.7%	-1.05	+4.3% (1.4)	-0.89	+9.2%	0.85	-21%	+10.5%	0.71	+10.3%

Monthly, local currency (World in USD). L/S = equal-weighted calmest stocks (lowest 252-day volatility) minus most volatile stocks, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

Low volatility on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjerner · Project 10 factsheets

2. Strategy description

Low-volatility stocks earn about the same or more than high-volatility stocks, so their risk-adjusted return is far higher, contradicting the CAPM. Blitz & van Vliet (2007) documented it globally for 1986–2006; Ang, Hodrick, Xing & Zhang (2006) for idiosyncratic volatility.

Why it might work. Leverage constraints push investors into high-beta, high-volatility stocks (Black 1972; Frazzini & Pedersen 2014); benchmarked managers cannot exploit it without tracking error (Baker, Bradley & Wurgler 2011); lottery preferences (Bali, Cakici & Whitelaw 2011).

Rule	Original (Haugen & Heins (1975); Blitz & van Vliet (2007))	This factsheet
Signal	Past 1–3 year volatility (weekly or daily returns)	252-day volatility of daily total returns
Portfolio	Decile or quintile; often long-only low-vol	Extreme quantile L/S and the low-vol leg long-only, monthly
Weighting	Equal or value weighted	Equal weighted

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP ccloseadj (total return), 805 tickers incl. delisted	Point-in-time: the S&P 500 members at each month-end (Sharadar add/remove history; corrected 2 Oct 2026)	~503	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as run_wiki_pit.py / run_eu_pit.py), with a short list of verified ticker renames (e.g. NZYM-B → NSIS-B, WDH → DEMANT, DAI → MBG). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-

quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

4. Signal creation

$VOL_{i,t}$ = standard deviation of daily returns over the 252 trading days to month-end t (at least 200 days). Signals use only information up to month-end t and are traded in month $t+1$. Eligibility: in the point-in-time universe on any of the last five trading days of month t .

5. Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50 ; fixed per universe).
- **Long/short** = calmest stocks (lowest 252-day volatility) minus most volatile stocks; **long-only** = calmest stocks (lowest 252-day volatility); benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

6. Performance in detail

US (S&P 500 members, point-in-time), median 502 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-8.7%	-11.6%	25.9%	-0.34 [-0.79, 0.14]	-0.55	-84%	45%	-1.36	-1.01	+4.6% (1.1)	19.7%
Long-only (net)	+10.6%	+10.4%	12.0%	0.89 [0.51, 1.34]	1.41	-21%	64%	5.13	0.57	+3.2% (1.7)	6.9%
Calmest stocks (gross)	+10.9%	+10.7%	12.0%	0.91 [0.53, 1.36]	1.45	-21%	64%	5.25	0.57	+3.4% (1.9)	6.9%
Most volatile stocks (gross)	+18.0%	+14.9%	28.6%	0.63 [0.20, 1.11]	0.89	-44%	61%	2.72	1.58	-2.8% (-0.9)	15.0%
EW universe	+13.1%	+12.6%	15.5%	0.85 [0.41, 1.42]	1.29	-28%	67%	4.20	1.00	–	9.2%
Size-weighted universe	+14.9%	+14.9%	14.2%	1.05 [0.60, 1.61]	1.73	-24%	69%	5.29	0.86	+3.7% (2.4)	8.3%

EU (11 national blue-chip indices, point-in-time), median 264 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+1.6%	-0.4%	19.9%	0.08 [-0.45, 0.68]	-0.03	-58%	52%	0.28	-0.87	+10.9% (2.4)	13.1%
Long-only (net)	+11.0%	+11.0%	10.3%	1.07 [0.51, 1.76]	1.69	-21%	67%	4.53	0.56	+5.0% (2.6)	6.3%
Calmost stocks (gross)	+11.3%	+11.3%	10.3%	1.10 [0.54, 1.79]	1.74	-21%	68%	4.64	0.56	+5.2% (2.8)	6.2%
Most volatile stocks (gross)	+8.2%	+5.7%	23.4%	0.35 [-0.17, 0.87]	0.40	-51%	53%	1.27	1.43	-7.1% (-2.1)	11.7%
EW universe	+10.8%	+10.2%	14.4%	0.75 [0.29, 1.30]	1.14	-26%	63%	3.20	1.00	–	8.4%
Size-weighted universe	+10.6%	+10.2%	13.7%	0.77 [0.32, 1.29]	1.21	-25%	58%	3.37	0.92	+0.8% (0.6)	7.9%

UK (FTSE 350, point-in-time), median 326 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-6.0%	-8.2%	21.5%	-0.28 [-0.82, 0.30]	-0.46	-79%	52%	-0.96	-0.92	+2.8% (0.6)	16.5%
Long-only (net)	+8.4%	+8.3%	9.1%	0.93 [0.41, 1.58]	1.37	-20%	64%	4.38	0.57	+2.9% (2.8)	6.1%
Calmost stocks (gross)	+8.6%	+8.5%	9.1%	0.94 [0.43, 1.60]	1.40	-20%	64%	4.48	0.57	+3.1% (3.0)	6.0%
Most volatile stocks (gross)	+13.4%	+10.6%	26.2%	0.51 [0.01, 1.06]	0.70	-46%	53%	1.87	1.50	-0.9% (-0.2)	13.8%
EW universe	+9.6%	+8.9%	14.2%	0.68 [0.19, 1.28]	0.95	-30%	58%	2.87	1.00	–	8.8%
Size-weighted universe	+8.4%	+7.8%	12.9%	0.65 [0.16, 1.25]	0.90	-28%	62%	2.68	0.84	+0.3% (0.2)	7.8%

DK (OMXC25, point-in-time), median 20 names, 3 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-1.1%	-2.7%	18.1%	-0.06 [-0.66, 0.50]	-0.20	-57%	51%	-0.21	-0.66	+7.2% (1.8)	11.1%
Long-only (net)	+13.3%	+13.3%	12.4%	1.07 [0.58, 1.58]	1.85	-19%	63%	4.53	0.68	+4.8% (2.4)	6.5%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Calmeest stocks (gross)	+13.5%	+13.5%	12.4%	1.08 [0.60, 1.60]	1.89	-19%	63%	4.60	0.68	+5.0% (2.5)	6.5%
Most volatile stocks (gross)	+13.5%	+11.6%	22.4%	0.60 [0.05, 1.25]	0.82	-46%	63%	2.18	1.34	-3.3% (-1.3)	13.3%
EW universe	+12.6%	+12.0%	15.2%	0.83 [0.31, 1.44]	1.27	-33%	66%	3.31	1.00	–	8.8%
Size-weighted universe	+11.1%	+10.2%	16.4%	0.68 [0.11, 1.33]	0.94	-40%	62%	2.55	0.93	-0.6% (-0.2)	10.7%

SCANDI (OMXC25 + OMXS30 + OMXH25, point-in-time), median 70 names, 5 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-3.1%	-4.5%	16.9%	-0.19 [-0.79, 0.38]	-0.35	-71%	49%	-0.63	-0.74	+6.1% (1.7)	10.1%
Long-only (net)	+12.2%	+12.3%	10.8%	1.13 [0.65, 1.70]	1.89	-15%	67%	5.05	0.66	+3.9% (2.6)	6.4%
Calmeest stocks (gross)	+12.4%	+12.5%	10.8%	1.15 [0.67, 1.73]	1.93	-15%	67%	5.15	0.66	+4.1% (2.7)	6.4%
Most volatile stocks (gross)	+14.2%	+12.5%	21.9%	0.65 [0.10, 1.28]	0.92	-45%	61%	2.35	1.39	-3.3% (-1.4)	12.4%
EW universe	+12.6%	+12.3%	14.2%	0.89 [0.37, 1.51]	1.38	-29%	66%	3.63	1.00	–	8.4%
Size-weighted universe	+11.4%	+11.0%	13.5%	0.84 [0.37, 1.41]	1.32	-26%	63%	3.59	0.90	-0.0% (-0.0)	8.1%

World (US + UK + EU, point-in-time), median 1088 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-5.7%	-7.7%	20.8%	-0.27 [-0.75, 0.22]	-0.46	-75%	47%	-1.05	-0.89	+4.3% (1.4)	14.9%
Long-only (net)	+9.4%	+9.2%	11.0%	0.85 [0.40, 1.43]	1.25	-21%	65%	4.48	0.60	+2.7% (2.1)	7.2%
Calmeest stocks (gross)	+9.6%	+9.4%	11.0%	0.87 [0.42, 1.46]	1.29	-21%	65%	4.59	0.60	+2.9% (2.3)	7.2%
Most volatile stocks (gross)	+13.9%	+11.0%	26.4%	0.53 [0.05, 1.06]	0.70	-43%	55%	2.10	1.49	-2.9% (-1.1)	13.8%
EW universe	+11.3%	+10.5%	16.0%	0.71 [0.25, 1.28]	1.02	-30%	61%	3.07	1.00	–	9.7%
Size-weighted universe	+11.0%	+10.3%	14.8%	0.74 [0.28, 1.31]	1.10	-28%	61%	3.19	0.91	+0.7% (0.8)	8.9%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2013	-13.0%	-25.9%	+4.8%	-30.1%	-14.3%	-9.4%
2014	+0.5%	+58.0%	+6.5%	-17.7%	+11.7%	+17.3%
2015	+22.6%	+22.0%	+3.6%	-12.5%	-10.5%	+16.5%
2016	-17.0%	-30.0%	-53.5%	-19.9%	-34.8%	-31.0%
2017	+0.3%	-11.0%	-23.8%	+18.6%	-5.0%	-17.2%
2018	+6.1%	+22.5%	-2.4%	-4.3%	+8.4%	+7.0%
2019	-8.8%	-16.7%	+18.5%	+33.8%	-17.7%	-8.3%
2020	-36.0%	-22.1%	-33.3%	-30.1%	-30.9%	-29.6%
2021	-18.4%	+11.4%	-1.4%	+10.7%	-7.1%	-9.1%
2022	+10.5%	+1.9%	+19.3%	+17.1%	+24.8%	+14.9%
2023	-25.9%	+0.7%	-12.2%	-11.0%	-8.5%	-16.2%
2024	+6.4%	+8.9%	+2.0%	+14.3%	+31.8%	+5.8%
2025	-28.8%	-0.1%	+2.1%	+26.5%	+22.5%	-12.7%
2026	-32.8%	+7.0%	-7.7%	-3.2%	-4.0%	-14.7%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+17.9%	+14.1%	+15.8%	+22.0%	+17.9%	+20.2%
2014	+18.4%	+19.4%	+8.6%	+9.3%	+19.8%	+10.8%
2015	+6.1%	+11.6%	+7.7%	+27.1%	+5.7%	+4.3%
2016	+10.7%	+2.1%	+8.6%	+0.6%	+13.4%	+2.7%
2017	+14.0%	+14.0%	+15.6%	+24.2%	+13.5%	+16.6%
2018	+3.5%	+5.6%	-4.6%	-13.9%	+2.1%	-2.8%
2019	+24.8%	+18.8%	+21.9%	+43.3%	+25.0%	+21.4%
2020	+7.4%	-12.6%	-2.3%	+17.9%	+1.2%	+3.9%
2021	+19.7%	+24.6%	+11.9%	+25.6%	+17.1%	+12.8%
2022	-2.1%	-7.8%	-9.6%	-9.0%	-4.3%	-7.8%
2023	-0.6%	+15.5%	+4.9%	+2.6%	+7.3%	+4.8%
2024	+14.0%	+13.6%	+9.8%	+10.2%	+12.4%	+11.2%
2025	+5.9%	+19.2%	+16.5%	+25.7%	+23.2%	+19.6%
2026	+4.9%	+18.5%	+12.6%	+8.7%	+16.4%	+11.3%

7. Trading record

The rebalance log per universe is in `results/xs_FS04_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	163	51 / 50	10%	9%	45%	-29.9%	2020-11
EU	163	27 / 26	10%	10%	52%	-29.1%	2020-11
UK	163	33 / 32	8%	10%	52%	-33.7%	2020-11
DK	163	7 / 6	9%	8%	51%	-14.0%	2020-11
SCANDI	163	14 / 14	9%	9%	49%	-13.3%	2020-11
World	163	109 / 108	9%	9%	47%	-28.4%	2020-11

Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: calmest stocks	Short: most volatile stocks
US	BRK.B, ATO, REG, DUK, FE, WEC, EVRG, O	MRNA, SNDK, LITE, SMCI, COHR, MU, MRVL, WDC
EU	IVG.MI, OBEL.BR, INGA.BR, IBE.MC, TRYG.CO, SRG.MI, SAMPO.HE, LOG.MC	NYR.BR, AVIO.MI, ZEAL.CO, ORSTED.CO, NOKIA.HE, BESI.AS, STMMI.MI, STMPA.PA
UK	CGT.L, PNL.L, TFIF.L, RICA.L, SAIN.L, BHMG.L, ALW.L, MYI.L	CWR.L, RPI.L, GDWN.L, ROR.L, TRST.L, SPI.L, SSIT.L, OCDO.L
DK	TRYG.CO, JYSK.CO, NDA-DK.CO, DANSKE.CO, ISS.CO, CARL-B.CO, NSIS-B.CO, COLO-B.CO	ZEAL.CO, ORSTED.CO, NOVO-B.CO, GN.CO, VWS.CO, PNDORA.CO, MAERSK-B.CO, AMBU-B.CO
SCANDI	TRYG.CO, SAMPO.HE, SHB-A.ST, INVE-B.ST, SWED-A.ST, NDA-SE.ST, SEB-A.ST, KESKOB.HE	ZEAL.CO, ORSTED.CO, NOKIA.HE, QTCOM.HE, NOVO-B.CO, GN.CO, BOL.ST, VWS.CO

8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	-6.8%	-3.06	0.86	value -0.98 (-11.0), low_risk +2.39 (+16.0), short_term_reversal -0.53 (-2.6)

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
EU	French Europe 5F + WML	-1.4%	-0.29	0.53	Mkt-RF -0.51 (-5.7), RMW +1.32 (+4.4), WML +0.47 (+2.9)
UK	JKP GBR 7 themes	-11.5%	-2.51	0.70	value +0.45 (+2.0), momentum +0.37 (+2.4), low_risk +2.13 (+8.6), quality +0.81 (+2.9)
DK	JKP DNK 7 themes	+0.4%	0.12	0.63	momentum +0.22 (+2.1), low_risk +1.38 (+9.8)
SCANDI	French Europe 5F + WML	-5.3%	-1.57	0.34	Mkt-RF -0.41 (-4.8), SMB -0.60 (-3.5), HML +0.37 (+2.1), WML +0.43 (+3.7)
World	JKP World 7 themes	-5.3%	-1.52	0.74	value -0.45 (-2.7), low_risk +1.66 (+9.9), short_term_reversal -0.72 (-2.6)

Long-only

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	+1.9%	1.39	0.82	mkt +0.92 (+24.6), size -0.26 (-3.3), value -0.22 (-3.2), low_risk +0.95 (+9.6)
EU	French Europe 5F + WML	+4.4%	1.57	0.55	Mkt-RF +0.52 (+11.0), RMW +0.54 (+2.6), WML +0.15 (+2.6)
UK	JKP GBR 7 themes	+3.5%	2.03	0.62	mkt +0.40 (+9.5), momentum +0.17 (+2.6), quality +0.23 (+2.4)
DK	JKP DNK 7 themes	+7.3%	2.80	0.59	mkt +0.63 (+13.7), low_risk +0.23 (+2.9), quality +0.18 (+2.4)
SCANDI	French Europe 5F + WML	+7.7%	2.97	0.50	Mkt-RF +0.51 (+11.2)
World	JKP World 7 themes	+4.0%	2.15	0.74	mkt +0.68 (+13.8), low_risk +0.61 (+5.9)

9. Statistical verdict

- **Gate:** $|t| > 2.87$ (2 books × 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in 1 of the six; passes in 0 of 6.
- **L/S CAPM alpha:** +2.8% to +10.9%, $|t|$ past the gate in 0 of 6.
- **Long-only:** Sharpe above the equal-weight universe in all six; alpha +2.7% to +5.0%, passing in 0 of 6.
- **Probabilistic Sharpe** of the L/S, $P(\text{true Sharpe} > 0)$: US 0.10, EU 0.62, UK 0.14, DK 0.41, SCANDI 0.25, World 0.15.
- **Publication decay:** gross L/S -8.2% to +2.0% a year in 2013–19 against -15.8% to +5.1% in 2020–26.

10. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate $p < 0.0020$.

10.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	+1.3%	-0.94	-12.3%	+13.6%	-0.5%	+0.4%	+1.0%	20%	2013-05 -13%, 2016-04 -12%, 2015-04 -9%
EU	+2.0%	-1.00	-10.7%	+12.7%	-0.5%	+1.5%	+0.5%	19%	2015-02 -13%, 2016-03 -12%, 2016-10 -11%
UK	-7.9%	-0.74	-8.6%	+0.7%	-0.4%	-1.3%	-6.5%	16%	2016-04 -18%, 2016-02 -17%, 2016-06 -11%
DK	-4.9%	-0.53	-8.1%	+3.2%	-0.4%	-0.5%	-4.4%	18%	2013-05 -10%, 2013-02 -9%, 2016-04 -8%
SCANDI	-8.2%	-0.70	-10.7%	+2.4%	-0.4%	-1.6%	-6.6%	18%	2019-06 -10%, 2016-06 -10%, 2016-11 -9%
World	-2.5%	-0.86	-9.3%	+6.8%	-0.4%	-0.6%	-2.0%	18%	2016-04 -13%, 2016-03 -9%, 2015-02 -8%

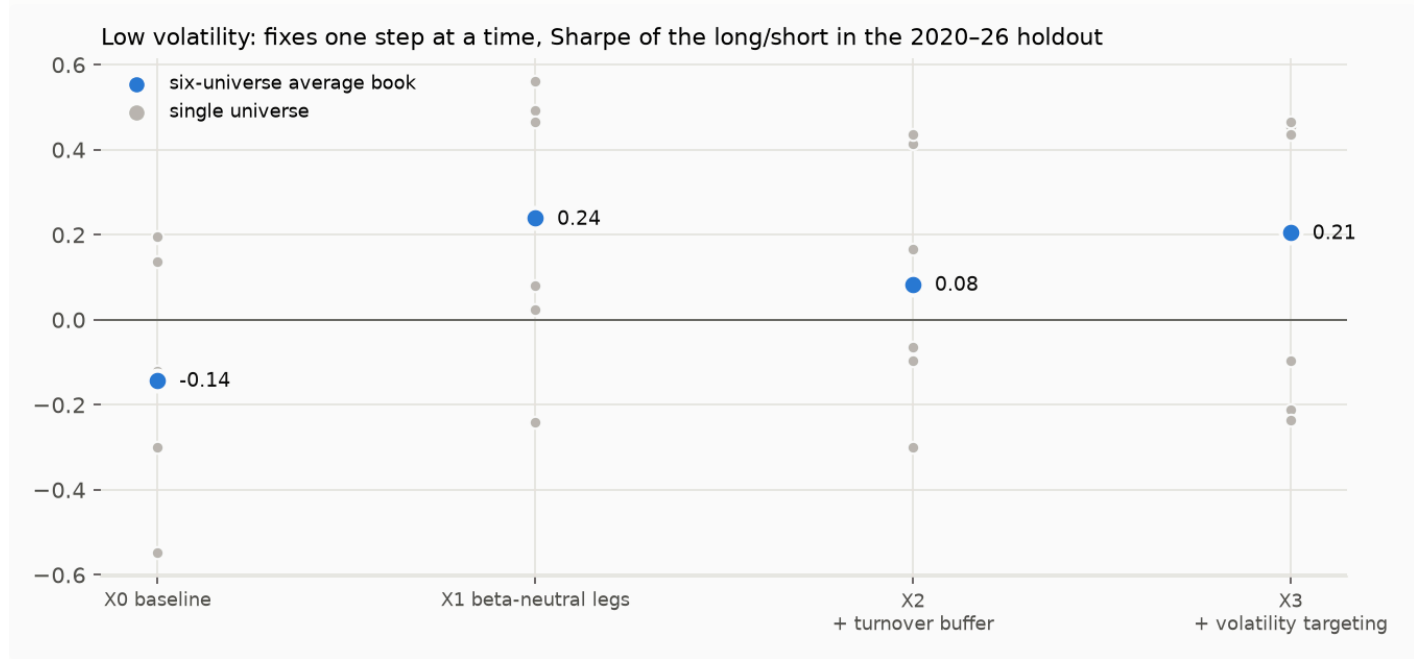
On average across the six universes the gross spread was -3.4% a year, of which the market exposure (beta -0.79) contributed -10.0%; the beta-adjusted spread (CAPM alpha) was +6.6%. Costs took 0.4% a year at 18% monthly turnover across both legs. The long leg beat the universe by -0.4% and the short leg lagged it by -3.0% a year, so most of the spread comes from the long side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +2.7%, +4.1%, +4.1% a year.

10.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	-0.29	–	-0.14	–	–	-2.9%	-49%	+6.3% (1.7)	0.4%
X1 beta-neutral legs	0.48	+1.04 (p 0.000)	0.24	+0.38 (p 0.071)	6 of 6	+3.6%	-29%	+2.9% (0.7)	0.6%
X2 + turnover buffer	0.50	+0.01 (p 0.495)	0.08	-0.15 (p 0.979)	0 of 6	+1.1%	-29%	+0.9% (0.3)	0.3%
X3 + volatility targeting	0.59	+0.14 (p 0.384)	0.21	+0.12 (p 0.206)	4 of 6	+2.0%	-22%	+1.9% (0.6)	0.2%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.



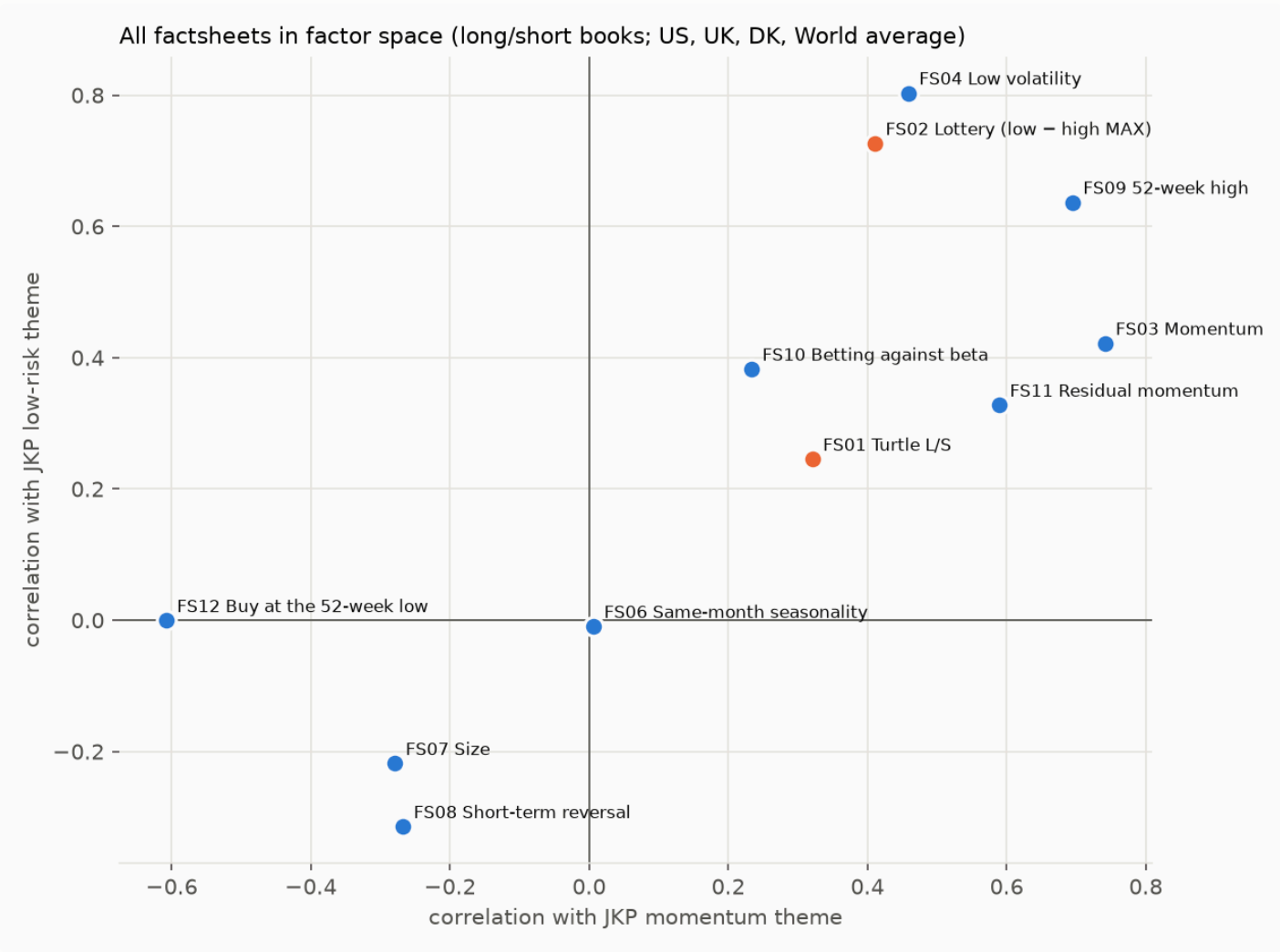
- **X1 beta-neutral legs:** +1.04 design, +0.38 holdout; helps in both windows but does not pass the gate.
- **X2 + turnover buffer:** +0.01 design, -0.15 holdout; helps in one window and hurts in the other: not reliable.
- **X3 + volatility targeting:** +0.14 design, +0.12 holdout; helps in both windows but does not pass the gate.

Where it ends: holdout Sharpe -0.14 → 0.21, return -2.9% → +2.0% a year (six-universe book).

11. Classification: where does it fit?

Dimension	Low volatility
Series family	Low risk
Academic style	Defensive / low-risk anomaly
Signal	Price only (daily total returns)
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month
Horizon and turnover	One month; 18% of the two legs replaced monthly
Market exposure	L/S beta -1.01 to -0.66
Payoff shape	Treynor–Mazuy γ -3.38 to -0.83 (t -7.9 to -0.8); worst 10% of market months +4.4% to +5.8% a month, best 10% -11.0% to -6.3%
Nearest factor theme	low_risk (4 of 6 universes), Mkt-RF (2 of 6 universes)

Dimension	Low volatility
Nearest library signals (returns)	Volatility (6 universes), Range (MAX–MIN) (5 universes), Beta (3 universes)
Economic rationale	Leverage constraints push investors into high-beta, high-volatility stocks (Black 1972; Frazzini & Pedersen 2014); benchmarked managers cannot exploit it without tracking error (Baker, Bradley & Wurgler 2011); lottery preferences (Bali, Cakici & Whitelaw 2011).
Publication	Haugen & Heins 1975; Ang et al. 2006; Blitz & van Vliet 2007; widely sold as min-vol ETFs since 2011
Where it fits	In the low risk cluster of the library; with some information beyond its neighbours.

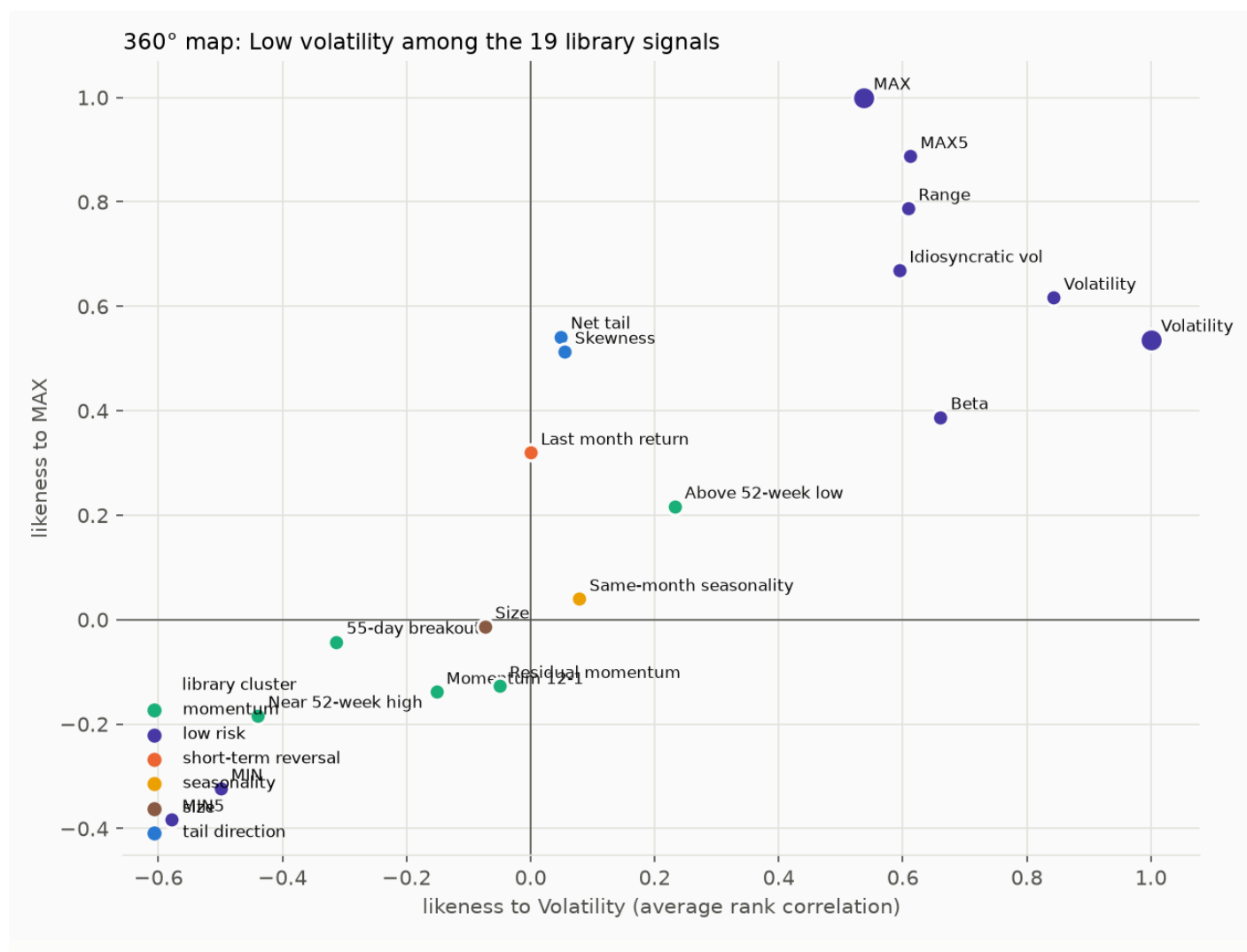


12. 360° view

The signal among the 19 price signals of the shared library (code/library.py), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **MAX**.

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t ≥ +2 / ≤ -2
Volatility (252 days) (this strategy)	low risk	+1.00	+1.00	+2.4%	-12.9% to -4.8%	0 / 3

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t ≥ +2 / ≤ −2
Volatility	low risk	+0.84	+0.92	+1.1%	-13.7% to -7.9%	0 / 5
Beta	low risk	+0.66	+0.86	+4.4%	-12.6% to -4.2%	0 / 4
Range (MAX−MIN)	low risk	+0.61	+0.84	-0.7%	-15.4% to -4.8%	0 / 4
MIN5	low risk	-0.58	-0.82	-0.9%	+2.3% to +13.7%	4 / 0
MAX5	low risk	+0.61	+0.79	+0.6%	-11.3% to -3.5%	0 / 2
MAX (best day) (mirror)	low risk	+0.54	+0.78	+0.1%	-13.3% to -2.0%	0 / 2
MIN (worst day)	low risk	-0.50	-0.78	-0.7%	+3.3% to +11.1%	4 / 0
Idiosyncratic vol	low risk	+0.59	+0.77	-3.6%	-15.5% to -4.8%	0 / 4
Near 52-week high	momentum	-0.44	-0.74	+3.7%	+9.3% to +22.3%	6 / 0
55-day breakout	momentum	-0.31	-0.63	+1.4%	+4.1% to +13.0%	4 / 0
Momentum 12-1	momentum	-0.15	-0.50	+7.2%	+4.3% to +19.4%	5 / 0
Size	size	-0.07	-0.33	+0.1%	-3.8% to +9.9%	1 / 0
Last month return	short-term reversal	+0.00	-0.28	+1.8%	-3.2% to +11.0%	2 / 0
Residual momentum	momentum	-0.05	-0.26	+6.4%	+8.0% to +12.2%	6 / 0
Same-month seasonality	seasonality	+0.08	-0.10	-1.9%	-9.4% to +6.6%	0 / 1
Skewness	tail direction	+0.06	+0.10	-0.4%	-6.7% to +3.9%	0 / 2
Above 52-week low	momentum	+0.23	+0.04	+7.5%	+5.2% to +13.6%	4 / 0
Net tail (MAX+MIN)	tail direction	+0.05	-0.01	-0.9%	-5.2% to +5.2%	0 / 0



Spanning. Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

Universe	Alpha after neighbours / yr	t	R ²	Neighbours with t ≥ 2
US	+1.2%	0.92	0.96	Volatility +0.51 (+5.8), Range +0.24 (+3.1), Beta +0.31 (+6.4)
EU	-0.6%	-0.27	0.88	Volatility +0.77 (+9.1), Beta +0.12 (+2.1)
UK	+5.4%	2.47	0.89	Volatility +0.46 (+5.3), Idiosyncratic vol +0.43 (+4.4), Range -0.27 (-2.5), Beta +0.28 (+5.8)
DK	-3.0%	-1.49	0.85	Beta +0.52 (+8.4), Volatility +0.29 (+4.7), Idiosyncratic vol +0.18 (+2.4)
SCANDI	-0.7%	-0.38	0.82	Volatility +0.64 (+7.9)
World	+2.6%	2.17	0.95	Volatility +0.56 (+6.7), Beta +0.33 (+7.1), Range -0.26 (-2.4), Idiosyncratic vol +0.32 (+4.3)

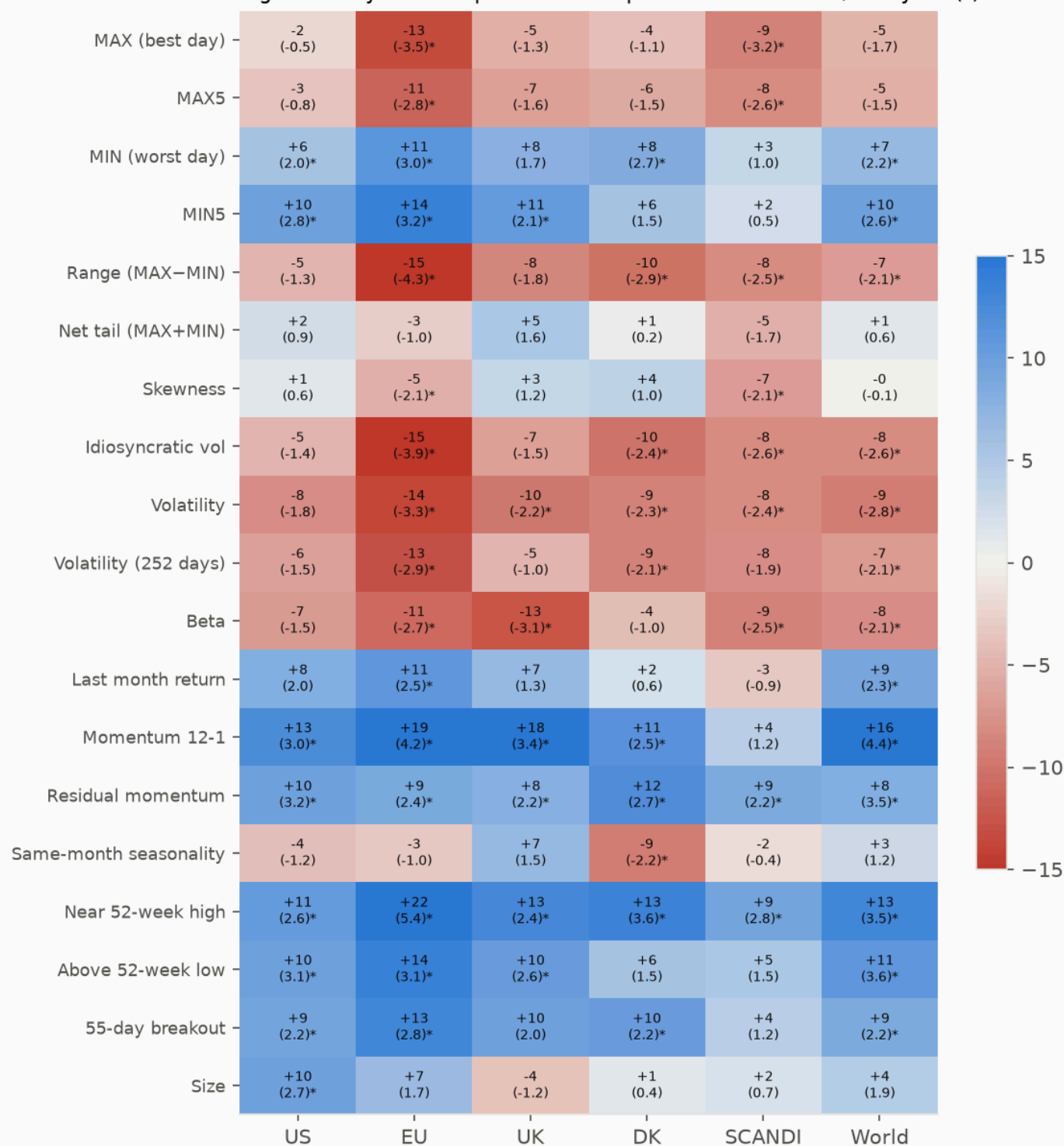
3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

low Volatility	11.3%	10.4%	8.1%
middle	10.2%	11.8%	10.8%
high Volatility	14.7%	14.2%	11.7%
	low MAX	middle	high MAX

Holding MAX fixed (down a column), moving from low to high Volatility changes the return by +3.4, +3.8 and +3.7 points a year; holding Volatility fixed (along a row), moving from low to high MAX changes it by -3.2, +0.6 and -3.0. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): +24% vs -36% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +2.4%, +2.7%, +4.1%, +4.1% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.

3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.
4. **Not a registered trial.** Descriptive; the fix ladder is counted inside FSO3–FS12 (25 trials).

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15. Reproduce

code/data.py → signals.py, signals2.py, library.py → xs.py + analyse_xs.py FS04 → build_xs.py FS04.
Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.